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PACTA + TRISK: Transition-Risk Analytics for Vietnamese Banks

Agenda

The Problem | Our Approach | Case Study | Next Steps

Our Solution: Three-Layer Compliance & Risk Stack

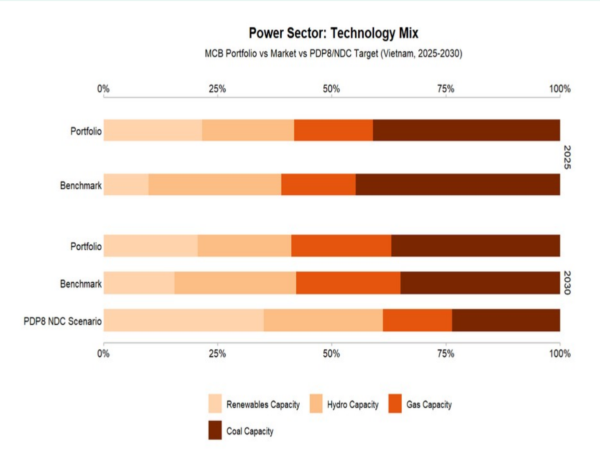
Layer	Tool	Purpose
1. Measure	PCAF	GHG inventory
	PACTA	Portfolio alignment
3. Stress	TRISK	Borrower stress
	IFRS S2	Board reporting

PACTA: Portfolio Alignment Assessment

Develop portfolio alignment reports using open-source R packages

Market-share method for power and automotive

SDA method for cement and steel emission intensity



43 loans, 25 trillion VND synthetic portfolio across power, automotive, cement, steel, and coal mining sectors

Custom PDP8 scenario from Vietnam Power Development Plan 8

Company-level results enable borrower-specific engagement

Scenario comparison: PDP8 vs IEA NZE vs Vietnam NDC

1,500+ institutions globally

Time to first results: 1-3 months

The Challenge: Decision 263 & Transition Risk

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3 SLL Support

THE CHALLENGE

Decisi ermal power, steel, and cement from 2025. Banks must assess borrower transition risk, but lack analytical tools to quantify portfolio alignment or borrower-level financial stress under Paris-aligned scenarios.

WHAT BANKS NEED

Vietr tack that measures portfolio alignment with PDP8/NDC targets, quantifies borrower-level NPV and PD impact under transition scenarios, and produces disclosure-ready outputs for IFRS S2 and TCFD.

TRISK: Borrower-Level Transition Stress Testing

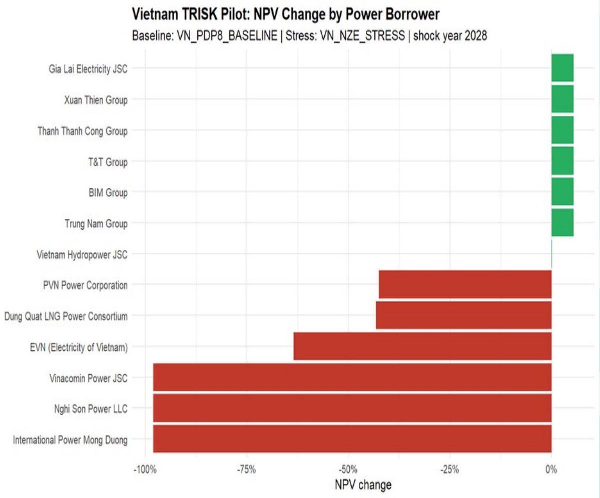
Quantify borrower-level NPV and PD impact under transition scenarios

Power: Coal-heavy borrowers show -12% to -18% NPV change

Cement: BF/BOF mills face -8% to -14% NPV impact

Steel: EAF routes outperform BF/BOF by 6-10pp

Priority ranking: alignment gap + NPV change + exposure



Covers all three Decision 263 sectors

Open-source R package (trisk.model) on CRAN

Engagement Offer & 2026 Roadmap

Phase 1: Data intake and loanbook normalization (BYOL)

Phase 2: PACTA portfolio alignment with real borrower data

Phase 3: TRISK stress testing and borrower ranking

Phase 4: IFRS S2 disclosure package and board-ready report

Timeline: 6-9 months from data receipt to final deliverables.

All analytics are open-source with no licensing fees.

We welcome the opportunity to discuss how this analytics stack can support your bank's transition-risk management and Decision 263 compliance.

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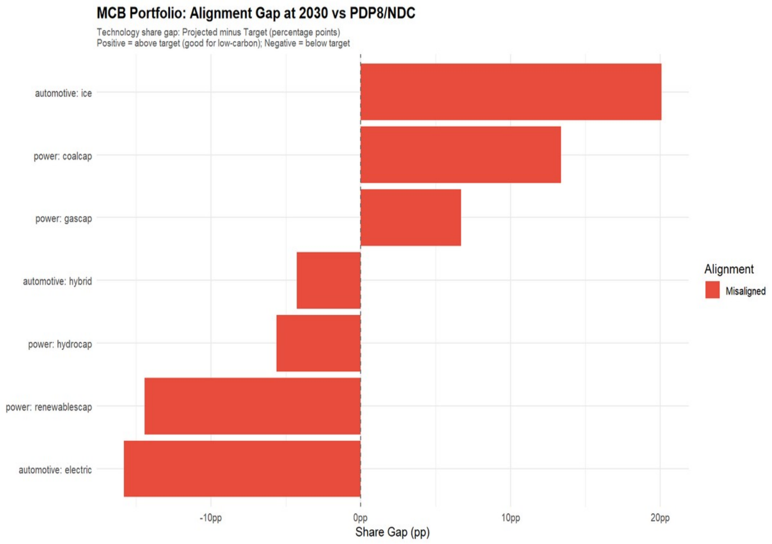
Case Study: Vietnam Bank Demo - Power Sector Alignment

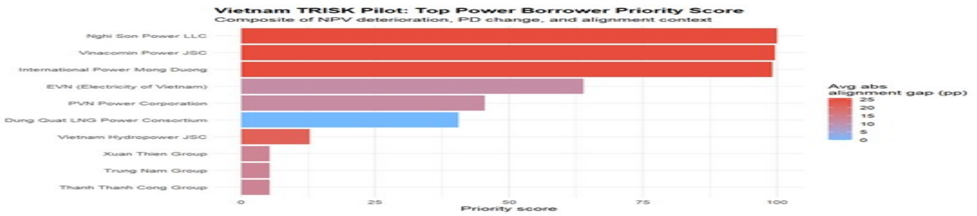
Metric	Value
Portfolio	43 loans, ~$1B USD synthetic
Power Mix	Coal 28%, Gas 12%, Hydro 10%, Solar 8%, Wind 5%
PDP8 Target	Coal phase-down, renewables 3x by 2030
Gap	Coal +28% above target; renewables -15% below

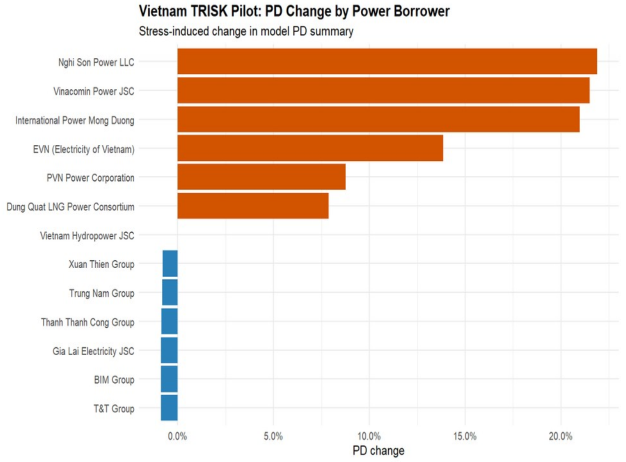


Case Study: TRISK Borrower Stress Rankings

TRISK ranks borrowers by financial stress severity. Coal-heavy names (Vinh Tan, Duyen Hai, Mong Duong) show highest NPV decline and PD increase. Renewable-focused borrowers (Trung Nam, BIM) benefit from transition tailwinds.

Illustrative / synthetic data - not actual portfolio results.





Priority Score (Top 10)

PD Change by Borrower

Case Study: Key Findings & Takeaways

Coal Stranded Risk

28% of portfolio coal capacity exceeds PDP8 phase-down trajectory. BOT plants face cliff-edge decline post-PPA expiry (~2035).

Renewables Gap

Renewables buildout is 15% below 2030 target. Solar and wind capacity must triple to align with PDP8.

Borrower Heterogeneity

NPV impact ranges from -18% (coal-heavy) to +8% (renewable-focused). One-size engagement fails.

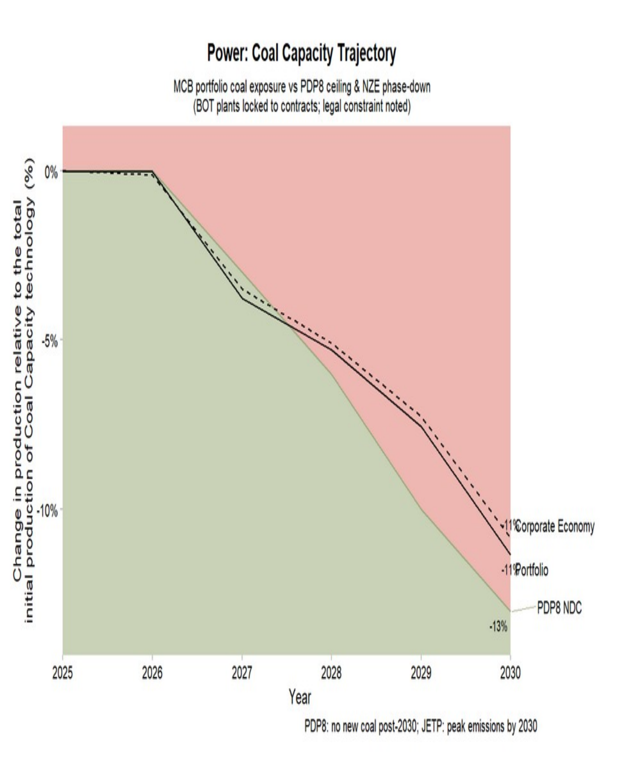
Decision 263 Readiness

Three-layer stack (PCAF, PACTA, TRISK) addresses all Decision 263 requirements: inventory, quotas, and reduction plans.

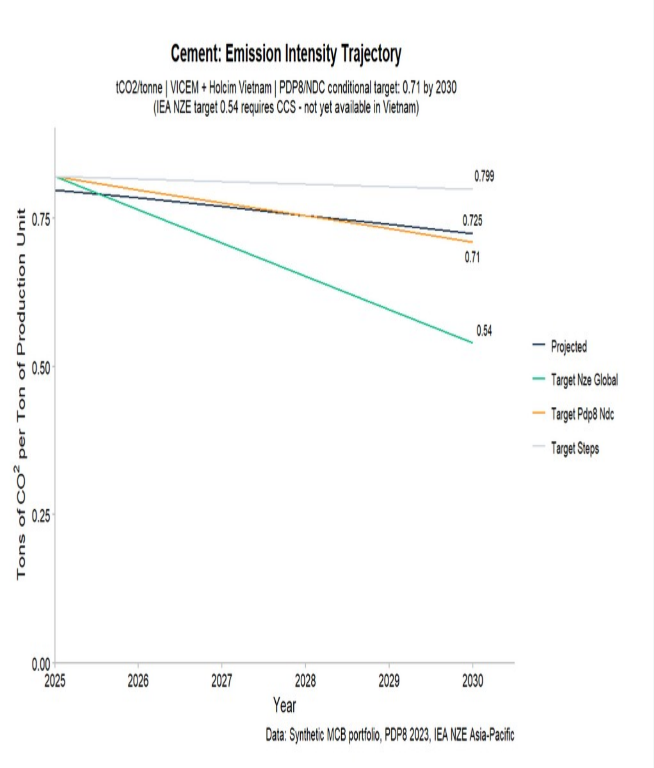
Interactive Exploration

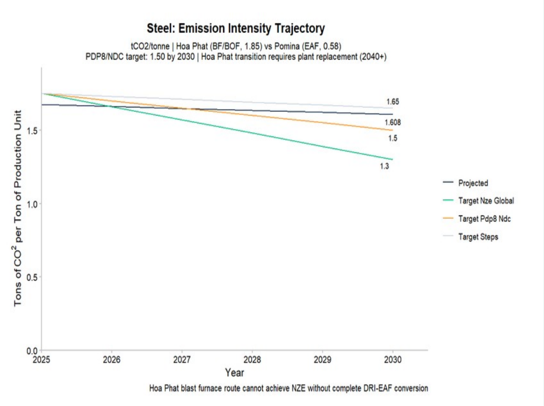
Live dashboard with Scenario Builder lets bank teams explore levers in real time. Downloadable HTML reports and CSV exports.

Compliance Value: Decision 263 / TCFD / ISSB



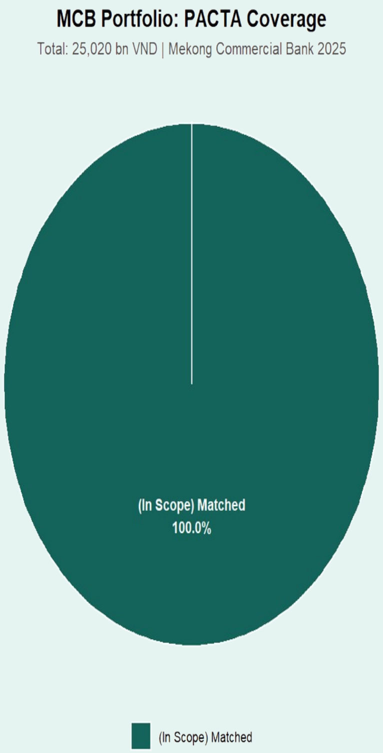
Coal trajectory vs. PDP8 phase-down pathway

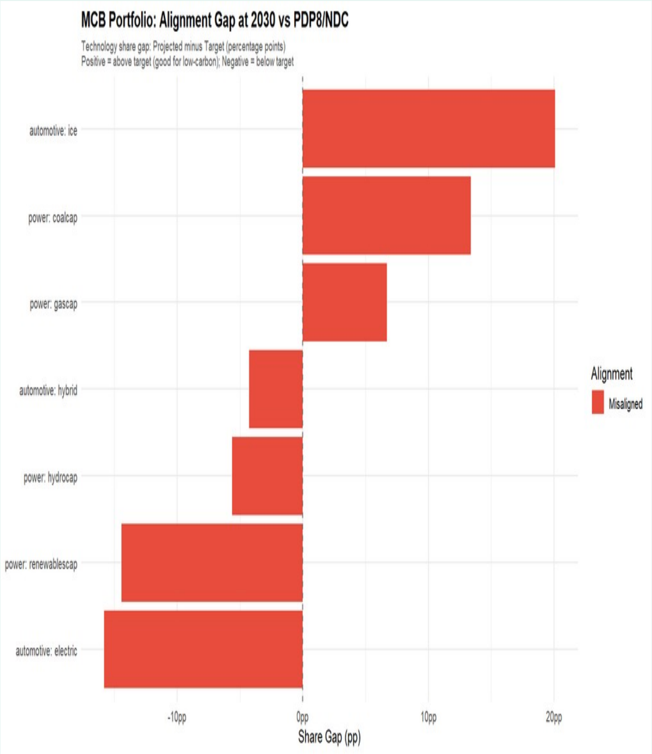




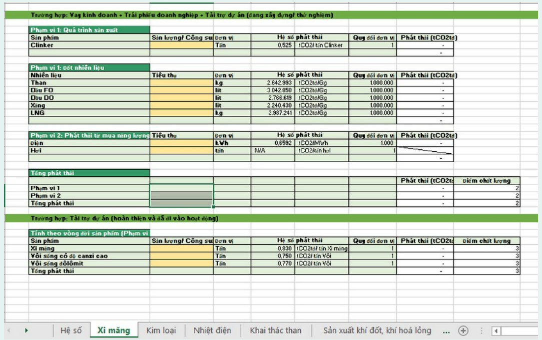
Cement & Steel SDA: Emission intensity vs. 2030 NDC targets

Live Dashboard: Interactive Portfolio Exploration





TRISK Risk: Borrower stress rankings with sensitivity controls

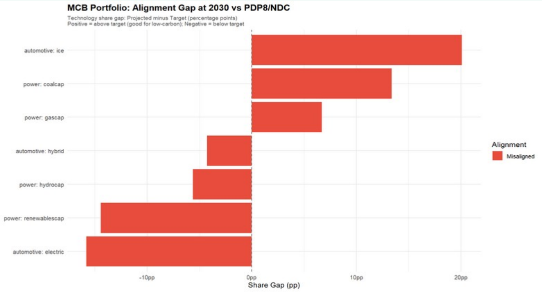


Trường hợp: Vay kinh doanh • Trái phiếu doanh nghiệp • Tài trợ dự án (đang xây dựng/ thử nghiệm)						
Phạm vi 1: Quá trình sản xuất						
Sản phẩm	Sản lượng/ Công su	Đơn vị	Hệ số phát thải		Quy đổi đơn vị	Phát thải (tCO2tđ)
Clinker		Tấn	0,525	tCO2/ tấn Clinker	1	-
						-
Phạm vi 1: Đốt nhiên liệu						
Nhiên liệu	Tiêu thụ	Đơn vị	Hệ số phát thải		Quy đổi đơn vị	Phát thải (tCO2tđ)
Than		kg	2.642,993	tCO2tđ/Gg	1.000,000	-
Dầu FO		lit	3.042,850	tCO2tđ/Gg	1.000,000	-
Dầu DO		lit	2.766,619	tCO2tđ/Gg	1.000,000	-
Xăng		lit	2.240,430	tCO2tđ/Gg	1.000,000	-
LNG		kg	2.987,241	tCO2tđ/Gg	1.000,000	-
						-
Phạm vi 2: Phát thải từ mua năng lượng	Tiêu thụ	Đơn vị	Hệ số phát thải		Quy đổi đơn vị	Phát thải (tCO2tđ)
Điện		kWh	0,6592	tCO2/MWh	1.000	-
Hơi		tấn	N/A	tCO2/tấn hơi	1	
						-

Tổng phát thải	Phát thải (tCO2tđ)	Điểm chất lượng
Phạm vi 1	-	2
Phạm vi 2	-	2
Tổng phát thải	-	2

Trường hợp: Tài trợ dự án (hoàn thiện và đã đi vào hoạt động)							
Tính theo vòng đời sản phẩm (Phạm vi							
Sản phẩm	Sản lượng/ Công su	Đơn vị	Hệ số phát thải		Quy đổi đơn vị	Phát thải (tCO2tđ)	Điểm chất lượng
Xi măng		Tấn	0,830	tCO2tđ/ tấn Xi măng	1	-	3
Vôi sống có độ canxi cao		Tấn	0,750	tCO2/ tấn Vôi	1	-	3
Vôi sống đôlômit		Tấn	0,770	tCO2/ tấn Vôi	1	-	3
Tổng phát thải						-	3

Hệ số | Xi măng | Kim loại | Nhiệt điện | Khai thác than | Sản xuất khí đốt, khí hoá lỏng | ...



Scenario Builder: Drive the five TRISK levers in real time